Traffic Lights

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Description

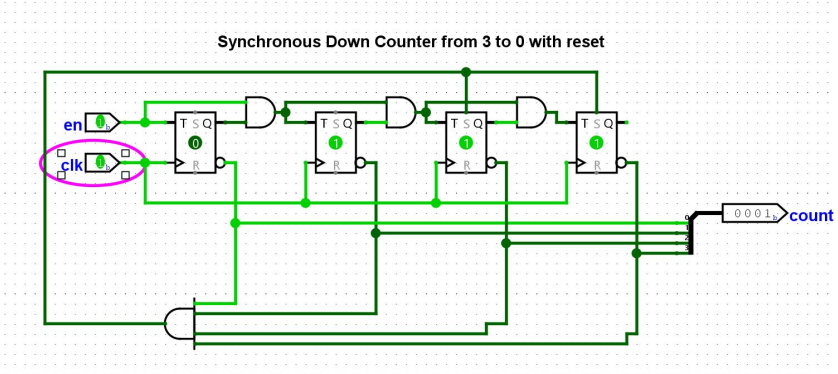
This final project aims to create two working traffic lights in logisim. Each light should interact in a specific way depending on which pedestrian is pushing the button.

Steps

1. **Start with Road A.** On power-up, Light A is green, Light B is red, and the display shows 03.
2. **Hold every color for 3 seconds.**Green and yellow each last exactly 3 s, then the sequence moves on.
3. **Let one road move at a time.**Whenever one light is green or yellow, the other light is red. The two lights are never green or yellow together.
4. **Remember every press.**A quick tap counts, even if the button is released long before the next clock edge. Without a press, the walk never happens.
5. **Finish the current green and yellow first.**A press never cuts a color short. The walk starts when the moving road's yellow ends, so a pedestrian waits at most 6 s.
6. **Give pedestrians 10 seconds.**Both lights stay red for exactly 10 s while the display counts down 10, 09, … 01.
7. **Ignore presses during the walk.**A press during the walk does not start a second walk.
8. **Resume on the other road.**After the walk, the road that did not just have its turn goes green. A walk after Light A's yellow continues with Light B's green, and a walk after Light B's yellow continues with Light A's green.

Pieces Being Used

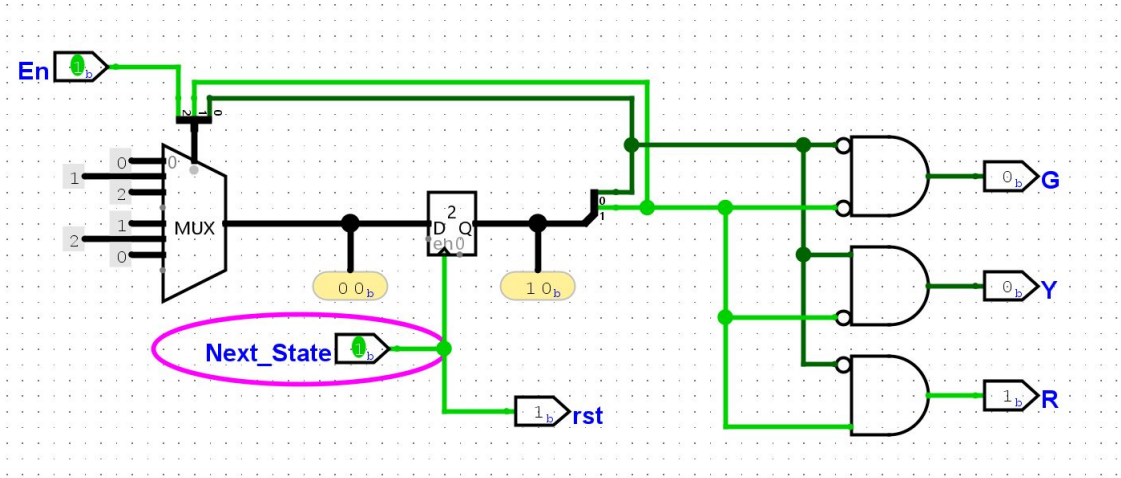
T-flip-flop count down from 3 to 0.



Right now this image is stopped at the number 3, after that it will go to 2, 1, 0, then reset.

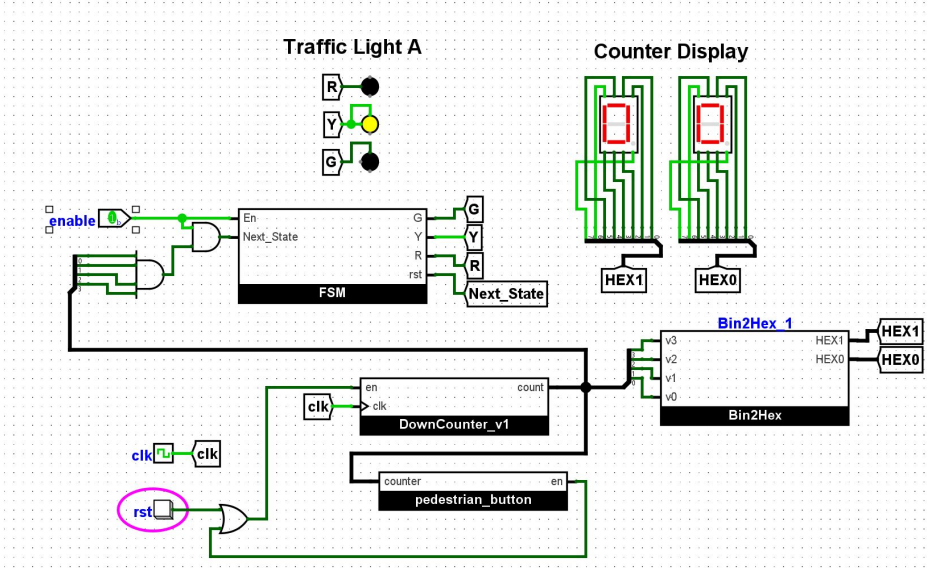
T-flip-flop count down from 10 to 0.

Finite State Machine Designs



This is the Final FSM machine built in Logisim. After you turn on the enable and the rst (reset) button, you can click the Next State button and move through green, yellow, and red while each output color waits for the timer to countdown from 2-0.

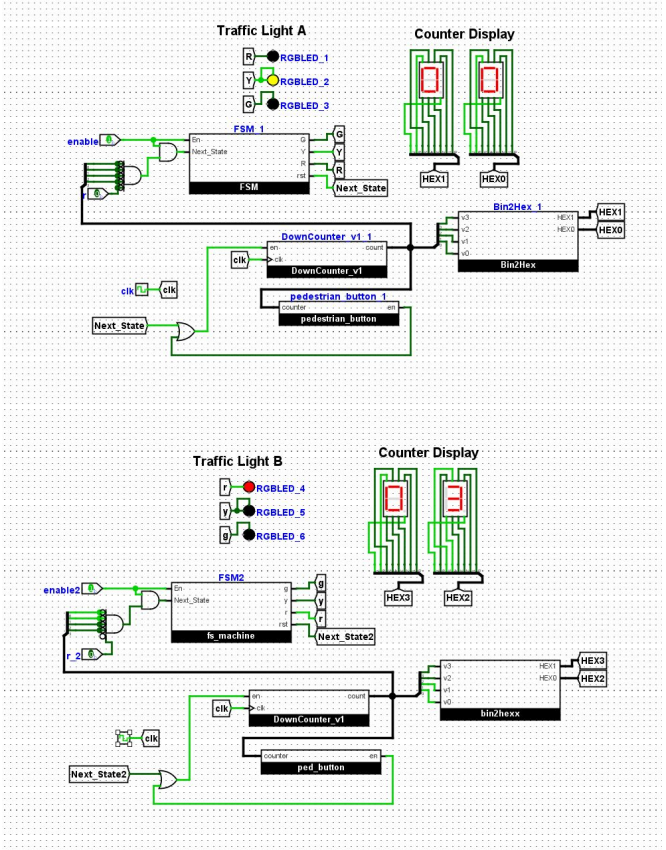
Connecting Everything Together

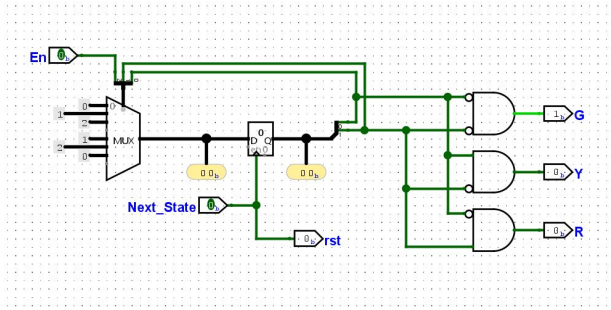


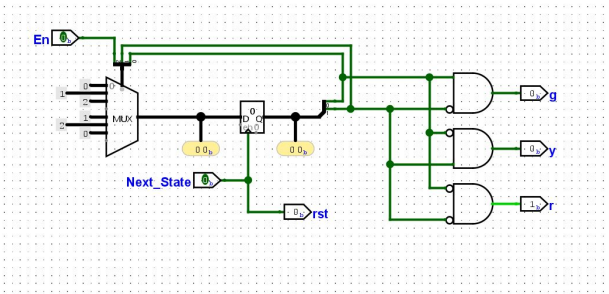
Here we connected the FSM to the rest of the logic. By connecting the FSM we are able to see the color of our traffic lights changing (green, yellow, red), this specific image is showing yellow! The Counter Display counts down from 0-3 once the reset button gets pushed. It then signals the FSM to wait the 3 seconds before changing to the next color (this is from the Next_State output). It moves to a different color once the reset button gets pushed.

Second light + 3 Counter

3-second countdown + Second Light



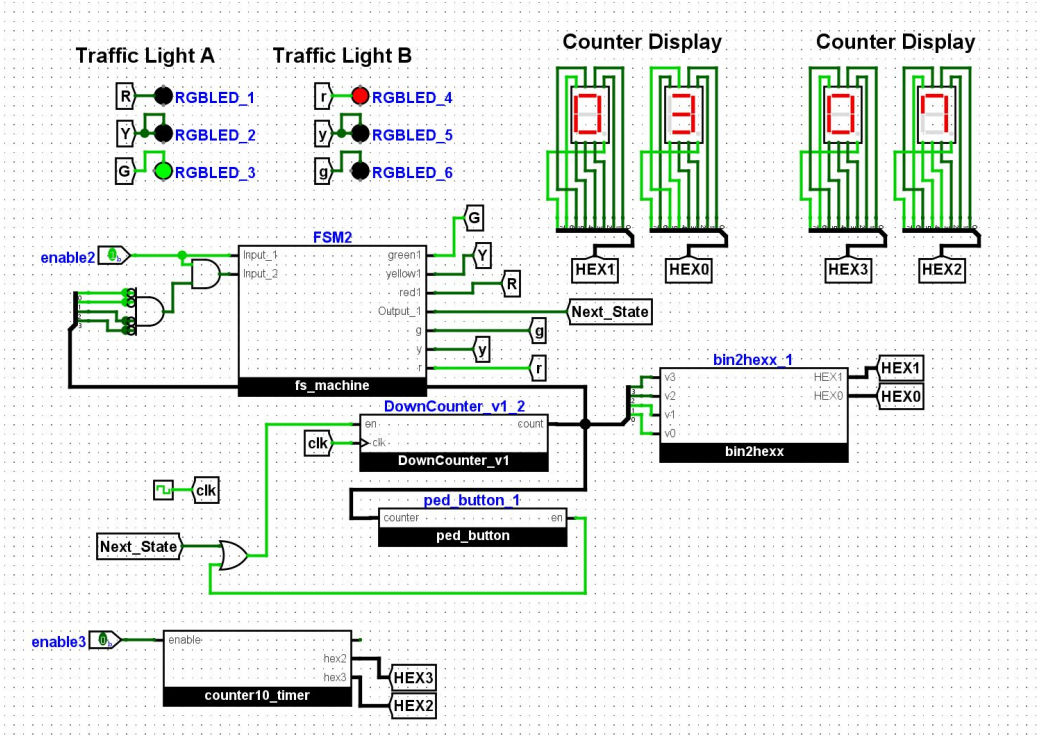




This is the 3 second counters for both pedestrians. I used the same logic from Traffic Light A to make Traffic Light B, I did have to change some variables like the HEX's on the counter display for it could work for this specific light.

I realized the lights weren't correct on Traffic Light B, it's supposed to be red when light A is green. To fix this I used the FSM for light B and did the inverse of the logic.

Traffic Lights + Counter Displays



Originally I thought making separate FSM machine and a bin2hexx for traffic light B, but that sourced a problem. It essentially created two brains, but I needed just one. So in this iteration I added all the traffic lights to one FSM machine, then inverted input2 so that when everything hits zero, they actually become one. This allows the program to keep the counter display going.

Reflection

I did end up getting stuck on the 10 second down counter. It's not connected to the main brain so it functions on it's own, which is an issue. To fix this, I think I can create a separate pedestrian button and connect it to the main logic so that it all functions together.

The colors are also not correlated properly, the yellow is supposed to switch when one light is about to turn red. I tried to invert the yellows in my FSM, but it didn't seem to work.